

Seven Deadly Sins: Digital Bodies, Emotional Duplication, and Generative Performance

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Abstract

This dissertation presents *Seven Sins*, a practice-based study that transforms bodily movement into generative emotional states through real-time visual computation. Rather than recognising or classifying emotion, the system reframes Kinect-captured motion into seven algorithmically defined identities—Pride, Envy, Wrath, Sloth, Greed, Gluttony, and Lust. Each identity is driven by a specific computational grammar, combining noise fields, displacement behaviours, colour modulation, and feedback dynamics to create distinct emotional atmospheres.

Through this framework, the project investigates how motion can operate as a catalyst for affective production, showing that the body can trigger emotional behaviour without functioning as a source of emotional data. It further examines how computational processes—particularly noise-driven motion, dynamic feedback, and colour-based modulation—shape the expressive qualities of digital embodiment. By positioning emotion as something that emerges from system dynamics rather than symbolic representation, the research contributes to ongoing discourse in generative art, digital performance, and affect-centred interaction design.

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Declaration

This thesis has been created by myself and has not been submitted in any previous application for any degree. The work in this thesis has been undertaken by myself except where otherwise stated.

1 Introduction

In contemporary digital environments, human movement is increasingly processed through computational systems that capture, analyse, and reinterpret bodily presence. Movements that would normally disappear in real time are instead translated into depth values, silhouettes, and other forms of measurable data. This dissertation examines how such data-driven renderings can be used not for prediction or behavioural categorisation, but as material for constructing new affective experiences.

Seven Sins is a practice-based investigation into the transformation of bodily motion into generative emotional states through real-time graphics. Instead of attempting to detect or classify emotion, the system maps the performer's movement into seven predefined computational affective configurations—Pride, Envy, Wrath, Sloth, Greed, Gluttony, and Lust. Each state operates through its own set of visual and behavioural rules, meaning that the same physical action can produce different outcomes depending on the active scene. In this way, the installation shifts interactivity away from straightforward cause – effect responses and toward a system in which movement becomes a source of emotional behaviour shaped by algorithms.

The research positions itself within discussions of digital embodiment, generative aesthetics, and posthuman theories concerned with the formation of the algorithmic self. Rather than treating the digital body as a direct reflection of the physical one, Seven Sins presents it as a space of multiplicity—something that can be fragmented, reshaped, or expanded into different affective forms. This perspective highlights the role of computation in constructing emotional atmospheres and intensities through visual processes rather than through symbolic representation.

The project is guided by the following research questions:

- How can generative visual systems construct emotional states from bodily motion without relying on recognition-based models?
- How do specific algorithmic processes—such as noise modulation, feedback accumulation, spatial displacement, and temporal latency—shape the affective qualities of generated visual states?
- What forms of digital embodiment emerge when a single body is articulated through multiple computational identities?

2 Related Work

Research at the intersection of motion capture, affective experience, and computational aesthetics provides the theoretical foundation for this project. Traditional affective computing, as established by Picard (1997), treats emotion as a recognisable signal that can be detected or classified from physical cues. Later developments in movement analysis (Glowinski et al., 2011; Camurri et al., 2003) further correlate parameters such as velocity, symmetry, and acceleration with discrete emotional states. While informative, these approaches position emotion as data rather than experience, framing the digital body as an input channel rather than an expressive entity.

Generative art offers an alternative model in which emotion emerges from system dynamics. Works by Akten (2017) and Anadol (2018) rely on stochastic behaviours—noise, turbulence, feedback loops—to produce affective atmospheres without explicit emotion classification. In these works, emotional tone emerges from the interaction of motion, textural change, and colour modulation within the system. Rather than conveying emotion through symbolic representation, the computational process shapes an atmosphere that is felt through its dynamics. Massumi's theory of affect as pre-cognitive intensity (2002) reinforces this framing, suggesting that emotion arises in the transitional space between sensation and interpretation.

Interactive digital artworks also demonstrate the potential of bodily input to generate immersive aesthetic experiences. For example, Random International's Rain Room (2012) uses presence rather than gesture to alter environmental conditions. Learning to See (Akten, 2017) reinterprets visual input through machine-learned textures, producing an image space shaped by the model's training rather than direct observation. These works prioritize experiential transformation over literal representation, aligning with the aims of Seven Sins.

Positioned within this landscape, Seven Sins diverges from recognition-based emotional systems and from reactive installations that mirror user behaviour. Instead, it treats motion as raw material for emotional construction. Through the combination of depth mapping, generative distortion, and feedback-driven behaviour, the project contributes to ongoing discourses in digital embodiment, generative aesthetics, and affect-centered interaction design.

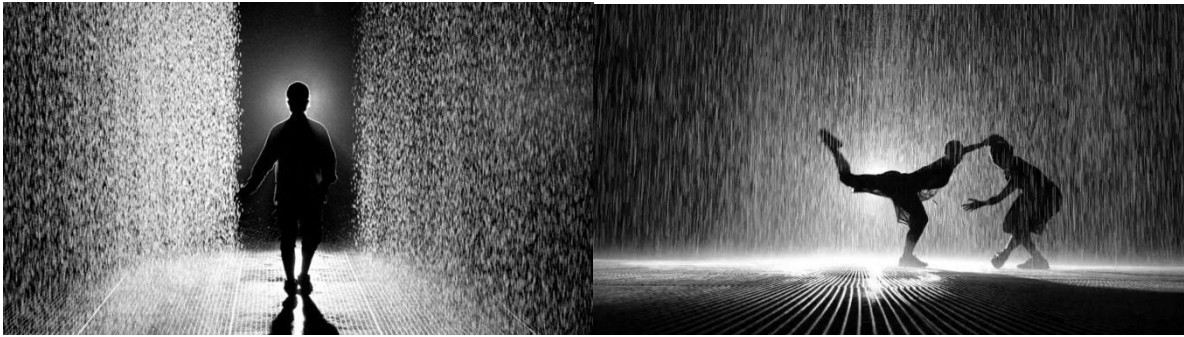


Figure 1: Image sourced from *Random International (2012)*, showing the *Rain Room* installation



Figure 2: Image sourced from *Memo Akten (2017) Learning to See*

3 Conceptual Framework

3.1 Seven Sins as Algorithmic Emotional Constructs

Each sin is treated as an algorithmically defined emotional construct shaped by the interaction of visual grammar and computational rules. Though all scenes share the same motion data, they diverge through transformations in colour, noise, delay, particle density, and feedback.

- Pride: emphasises centrality, structure, and ceremonial illumination, producing an atmosphere of grandeur and distance.
- Envy: manifests through duplication, splitting, and ghosted forms, generating a sense of self-other displacement.
- Wrath: evokes tension and eruption through unstable, high-intensity visual dynamics.
- Sloth: shapes an atmosphere of suspension through delayed responses, slow gradients, and desaturated space.
- Greed: expresses pressure, possession, and self-amplification through layers of visual accumulation and dominance.
- Gluttony: conveys expansion and excess through accumulating, overflowing visual structures.
- Lust: operates through softened transitions, glow, and fluid optical textures, emphasising tactility and attraction.

Rather than labelling emotions, these scenes construct emotional states through algorithmic manipulation. The performer becomes multiplied into seven digital identities, each reframing the same gesture within a different aesthetic and affective system.

3.2 Translating Affective Concepts into System Behaviour

Transforming emotional concepts into computational behaviour requires a structured translation method. Seven Sins achieves this through a shared set of visual primitives—feedback gain, delay, saturation, noise amplitude, and density accumulation—that operate as the foundational grammar for emotional construction.

Each sin modulates this grammar differently, forming not just stylistic variations but distinct emotion-generating systems. Thus, the technical design is itself part of the research methodology: a process in which emotion is formalised through algorithmic rules, enabling the digital body to function as an affective generator rather than an analytical object.

4 Methodology

This project adopts a practice-based research approach. The development of the system itself became a way of thinking through the artistic questions—particularly how movement can evolve into emotion within a computational environment. Rather than beginning with a fixed specification or an engineering blueprint, the project evolved through cycles of experimentation, visual testing, and conceptual adjustment. These cycles shaped not only the technical system but also the emotional framework that defines the seven scenes.

4.1 Practice-Based Research Approach

This project is grounded in a practice-based research framework, in which artistic creation is understood as a form of inquiry (Candy & Edmonds, 2018). Rather than beginning from predefined emotional categories, the research progressed through direct engagement with the materials of the system—depth maps, noise fields, motion-driven deformation, and feedback accumulation. Following established practice-based methods, these technical components were approached as creative materials whose behaviour had to be discovered through making, rather than as engineering problems to be solved (Mäkelä, 2007).

Early experiments were intentionally exploratory. I tested how silhouettes fractured under different noise frequencies, how extended feedback gradually increased visual “weight,” and how colour temperature altered the perceived tone of a scene. These trials were not designed to produce correct outcomes but to build an initial vocabulary of emotional tendencies—tension, heaviness, instability, dissolution—that would later guide the design of each sin.

4.2 Prototyping as Visual and Affective Exploration

The development process unfolded through a sequence of prototypes, each examining a specific aspect of the system: responsiveness to movement, clarity of visual behaviour, and the emotional readability of each sin. Rather than refining toward a single optimal configuration, each iteration explored how parameter adjustments—such as noise amplitude, feedback duration, or colour mapping—shifted the expressive character of a scene.

This iterative process served as a method for discovering the boundaries of the system. Prototypes that reacted too literally to motion or produced excessive visual complexity were not

treated as failures but as indicators of how emotional ambiguity could be calibrated. Through repeated cycles of testing and adjustment, the visual language of the seven sins gradually became more defined.

4.3 Motion-to-Emotion Mapping

A central component of the methodology was establishing how different types of movement could correspond to different emotional states. These mappings were not predefined but emerged gradually as the prototypes evolved. Slow drifting gestures consistently produced imagery associated with dissolution, while bursts of speed created forms that felt aggressive or disruptive. Micro-movement tended to trigger nervous or unstable patterns, whereas stable, centered poses created a sense of elevation or self-focus. Over time, these behavioural tendencies became codified into the emotional logic that differentiates the seven scenes.

The following matrix summarizes how each sin interprets motion and translates it into visual behaviour:

Sin	Motion Type	Visual Response	Participant Experience	Notes / Parameter Adjustments
Pride	Center-pose stability	Halo lighting, rhythmic expansion	Sense of centrality and structure	Gamma, Timer CHOP adjustment
Envy	Micro-movements	Flickering, green fractured textures	Sense of tension and instability	Noise amplitude, Difference TOP
Wrath	Velocity-sensitive	Red-green explosive gradients, jagged trails	Sense of emotional outburst and intensity	Skeleton velocity → Displace TOP
Sloth	Slow movement	Soft, melting boundaries, delayed response	Sense of sluggishness and inertia	Long Blur + Displace chain
Greed	Velocity-sensitive	Expanding gold-block accumulation	Sense of greed, continuous growth	Feedback + Tile Displace TOP
Gluttony	Large gestures	Swelling, feedback accumulation	Sense of excess and expansion	Noise + Blur + Feedback
Lust	Micro-movements	Soft gradients, slow pulsation	Sense of attraction and softness	Alpha blending + Timer CHOP

Diagram: *Seven Sins Interaction Matrix (Summary of Motion Mapping)*

4.4 Real-Time System Construction as Part of the Research Process

The construction of the real-time system functioned as an integral part of the research process

rather than a purely technical task. Throughout development, I experimented with how Kinect depth data, OSC transmission, and TouchDesigner’s visual networks interacted with one another, observing how each adjustment subtly changed the emotional behaviour of the scene. Technical choices were therefore guided by aesthetic intention: depth data shaped the presence and volume of the digital body; noise-based deformation introduced instability; feedback structures generated residue and accumulation; and delay operators produced heaviness, sluggishness, or suspension.

These components were not fixed from the outset. They shifted continuously as I tested different combinations—sometimes weakening feedback to achieve a cleaner silhouette, only to realise that this removed the emotional “weight” of the scene; other times reducing delay so that the system did not feel unresponsive. Through these cycles of trial and adjustment, each technical module gradually became tied to a specific emotional quality. The resulting pipeline was therefore not simply a mechanism for interaction, but a system whose expressive potential emerged through the interplay of computational behaviours.

4.5 Exploratory User Testing and System Adjustment

Before formal evaluation, the system was tested informally with a small number of participants to identify issues that were not apparent through solo experimentation. These sessions focused on practical concerns such as gesture clarity, sensitivity thresholds, and the stability of scene transitions. For example, rapid arm movements frequently caused unintended switching between sins, which informed the decision to simplify the gesture input to a single controlled hand-closure signal.

These informal encounters were not treated as data collection but as reflexive moments within the methodology, helping refine the interaction model and improve the emotional legibility of each scene. The adjustments made during this stage directly influenced the stability and readability of the final installation.

5 System Design & Implementation

This chapter outlines the technical design of Seven Sins, detailing how bodily movement is captured, processed, and transformed into real-time emotional visualisations. The system functions as an integrated “emotional computation engine,” in which motion data becomes the raw material for generating seven algorithmically distinct emotional states. The following sections describe the hardware configuration, software architecture, gesture-switching logic, and the shared computational principles that shape the project’s visual grammar.

5.1 Hardware Setup

The installation runs on a compact mobile workstation configured to sustain real-time generative rendering and low-latency motion processing. The system specifications are as follows:

- **Kinect v2 depth sensor**, operating with Kinect for Windows SDK 2.0, providing depth maps, skeletal tracking, hand-state detection, and gesture events at ≈ 30 fps.
- **NVIDIA GeForce RTX 3060 Ti (8 GB GDDR6)** GPU, TOP-level image processing, and GLSL fragment-shader execution.
- **Intel Core i7-12700H** (14 cores / 20 threads, base 2.3 GHz, turbo up to 4.7 GHz)
- **32 GB DDR4 RAM** and **1 TB NVMe SSD** for stable data throughput during live visual transformation.
- **Projection display** (1080p large-scale projector, ≥ 3500 lumens)
- **OSC-based communication layer** linking Kinect SDK $\rightarrow$ OSC $\rightarrow$ TouchDesigner with minimal network latency..

This configuration ensures that subtle bodily gestures captured by the Kinect sensor can be translated into continuous, high-resolution generative behaviour without compromising system stability or frame rate.

5.2 Software Architecture

The software system adopts a modular, end-to-end pipeline that organises motion data, computational processing, and visual synthesis into clear functional layers. This structure supports flexible experimentation across all seven emotional scenes and allows visual logic, colour grammars, and feedback behaviours to be adjusted independently.

5.2.1 Motion Data Pipeline

The core infrastructure follows a three-stage data flow:

Kinect for Windows SDK 2.0

Extracts depth maps, skeletal joints, joint velocity, hand-states, and gesture events at approximately 30 fps. These signals form the primary motion-state descriptors used across all scenes.

OSC Transmission Layer

Motion data is packed and transmitted to TouchDesigner via lightweight OSC channels, ensuring low-latency delivery of joint positions, velocities, and gesture flags.

TouchDesigner Processing Network

Inside TouchDesigner, the incoming data is reorganised through CHOP, TOP, and GLSL operators:

- CHOP subsystem: Kinect CHOP, Math CHOP, Timer CHOP
- TOP subsystem: Noise, Displace, Lookup, Ramp, Threshold, Blur, Edge Detect, Composite, Tile TOP
- GLSL shaders: Custom GLSL TOPs and lookup shaders (This multi-layered node system converts motion descriptors into visual parameters such as displacement intensity, texture behaviour, instability patterns, micro-movement response, colour transitions, and feedback accumulation.)

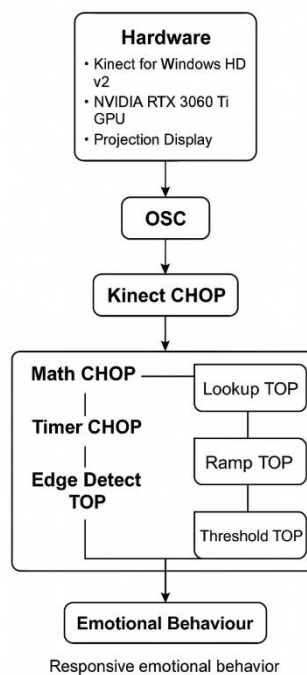


Figure 3: *Touchdesigner Processing Network Overview*

These parameters establish the computational basis for emotional transformations in each of the seven scenes, forming the system’s internal logic for translating bodily motion into generative affect.

5.2.2 Modular Scene Networks Architecture

The seven sins are implemented as parallel visual-processing “network islands” inside **a single TouchDesigner project file**.

Each island contains a complete image-processing chain responsible for transforming incoming motion data into its corresponding emotional behaviour. Every scene follows the same four-stage internal structure:

- **depth-silhouette extraction** (Kinect depth TOP → edge refinement → compositing)
- **noise-driven deformation** (Noise TOP, Displace TOP, GLSL distortion shaders)
- **colour mapping and emotional tonality** (Ramp TOP, Lookup TOP, Level TOP)
- **feedback-based temporal behaviour** (Feedback TOP chains, delay modulation, accumulation)

Although these stages share a common pipeline logic, the parameterisation and operator composition differ across scenes, producing seven distinct emotional outputs.

All seven scene networks run simultaneously within the same TouchDesigner project, organised as independent component containers (COMPs). Kinect-derived data channels—such as velocity, micro-movement, hand-state, and positional deviation—are broadcast globally through CHOP exports and OSC-in CHOPs, making the same motion data available to every scene.

A global **routing controller** selects which scene’s output is displayed. This controller switches the final TOP output by enabling or bypassing specific scene composites at the end of the render pipeline, ensuring seamless scene transitions without interrupting real-time processing.

5.2.3 Gesture Recognition and Scene Switching

Scene transitions are controlled via a dedicated gesture-recognition subsystem based on Kinect's hand-state detection. The system monitors the performer's right-hand fist state and processes it through:

- Select CHOP – extracts the right-hand fist state channel from the Kinect input stream
- Logic CHOP – converts the continuous hand-state values into a stable binary trigger signal
- Count CHOP – ensures that each detected fist gesture produces a single discrete transition event
- Null CHOP – serves as the final output node for signal clean-up and network organisation

Earlier prototypes assigned different gestures to each sin, but user testing revealed frequent accidental switching due to Kinect's sensitivity. Consolidating transitions into a single, deliberate fist gesture significantly improved reliability and preserved aesthetic continuity.

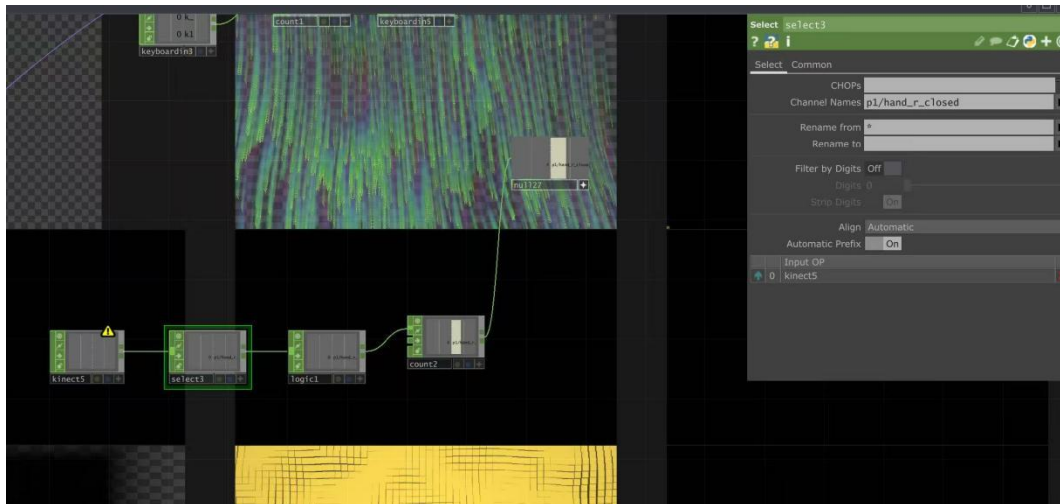


Figure 5: *Gesture-switching subsystem using Kinect right-hand fist detection*

5.2.4 Latency and Real-Time Performance

Because emotional meaning in Seven Sins emerges through motion-driven deformation, low latency is essential. The system maintains a responsive 50 – 60 fps through:

- compressed OSC transmission
- adaptive feedback regulation to avoid accumulation overload

- GPU-aware scaling of displacement and noise intensity
- timer-based synchronisation for rhythmic scenes (e.g., Pride, Lust)

These measures ensure smooth responsiveness even in computationally heavy scenes such as Wrath or Gluttony.

5.2.5 Unified TouchDesigner Layout

The entire installation runs within a single TouchDesigner project organised into three layers:

- Input Layer – handles Kinect skeleton, depth, and gesture CHOPs.
- Processing Layer – contains all distortion, noise, feedback, colour, and GLSL operators.
- Compositing Layer – routes, blends, and outputs the selected emotional scene.

This unified architecture keeps GPU resources compiled in memory, ensuring stable performance while reinforcing the conceptual model: a single digital organism containing multiple coexisting emotional behaviours.

5.3 Shared Visual Grammar

Despite their differences, all seven emotional scenes share a set of foundational computational primitives:

- **Feedback gain** → accumulation, persistence, excess
- **Time delay / lag** → inertia, heaviness, suspension
- **Saturation and colour mapping** → emotional temperature
- **Noise amplitude / turbulence** → instability, agitation
- **Particle / density accumulation** → growth, pressure, intensification

Each sin modulates these primitives in a distinct way, producing contrasting emotional identities while preserving internal coherence. This shared grammar allows Seven Sins to operate as a comparative emotional framework in which the same motion can generate radically different emotional worlds.

6 Technical Analysis of the Seven Sins

This chapter examines how each emotional state in Seven Sins is constructed through real-time computational behaviour. Although all scenes receive the same motion input from the Kinect pipeline, each sin transforms this input through a distinct combination of visual logic, deformation patterns, and motion-response strategies. The analysis below outlines the emotional rationale, system behaviour, and interactive characteristics of each sin.

6.1 Pride

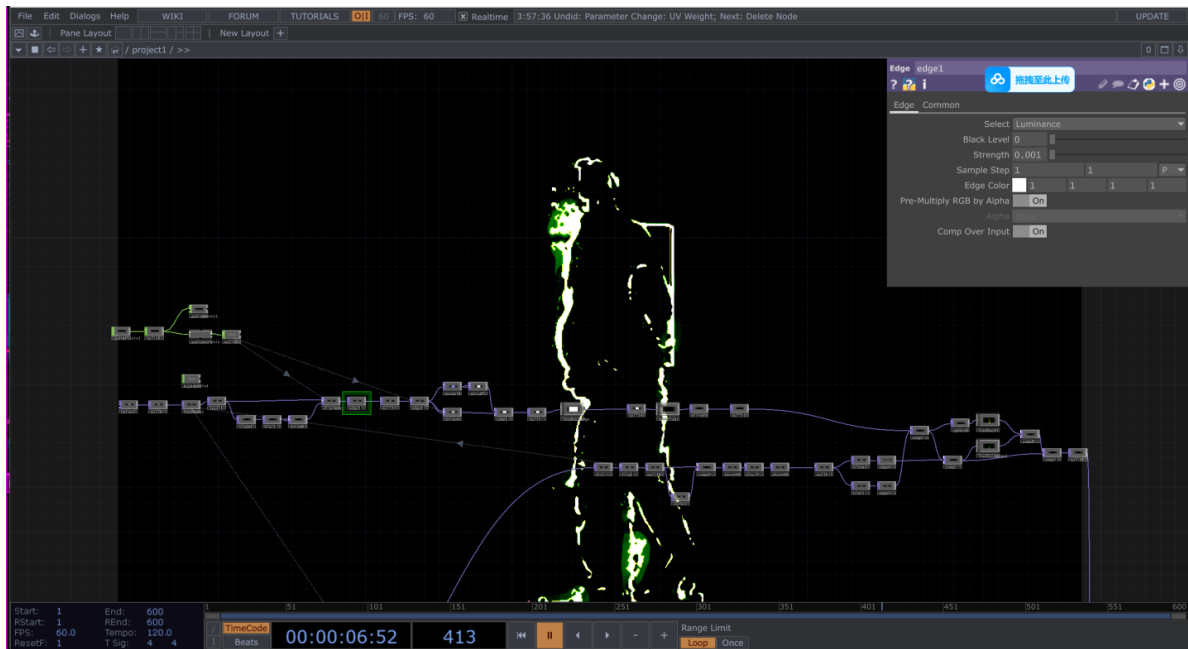


Figure 6: *TouchDesigner scene network for the sin Pride*

Emotional Logic

“Pride” is built around stability, clarity, and self-amplification. Rather than exaggerating motion, the scene elevates the body as a luminous central figure, reinforcing ideas of authority and heightened self-awareness.

System Behaviour

The Pride network is built as a system of structural amplification, where contour sharpness and centre-weighted luminance are used to construct an enhanced sense of self-presence. Kinect depth and skeleton data form the foundation of the processing pipeline. After initial background

suppression, the depth map is passed through an Edge Detect TOP with a near-maximal strength value, producing neon-like contours that magnify the outline of the body. This contour-enhancement layer establishes the dominant, self-highlighting visual language of Pride.

Following edge extraction, a Lookup + Ramp mapping converts depth Z-values into a gradient that intensifies toward the centre of the body, creating a strong halo effect that distinguishes the figure from its surroundings. A gamma-boosted Level TOP reinforces this luminance hierarchy, giving the impression of spotlight illumination and central authority. Unlike other sins that feature noise or unstable deformation, the Pride system is intentionally stable; its technical design avoids chaotic modulation to maintain the sense of poise and deliberate self-presentation.

To evoke ceremonial self-expansion, a Timer CHOP drives a gentle rhythmic scaling applied to the silhouette. This “breathing expansion” operates independently from performer movement, allowing the outline to subtly pulse at a controlled tempo. This rhythmic modulation enhances the symbolic qualities of pride: calm inflation, restrained power, and an aura that grows outward without losing form.

Collectively, the system prioritises structural clarity, controlled deformation, and a centred visual hierarchy. The pipeline is intentionally rigid and polished, engineering Pride as a technically sculpted visualization of exaggerated self-presence.

Motion Mapping

Pride responds most strongly to stillness and composure. When the performer maintains an upright, centered posture, the halo intensifies and the structural clarity sharpens. Minor drifting or lack of stability softens the luminance dominance, reducing contour precision. Motion therefore subtly moderates the visual system, but the emotional expression remains primarily tied to posture and centred balance rather than high-velocity movement.

6.2 Envy

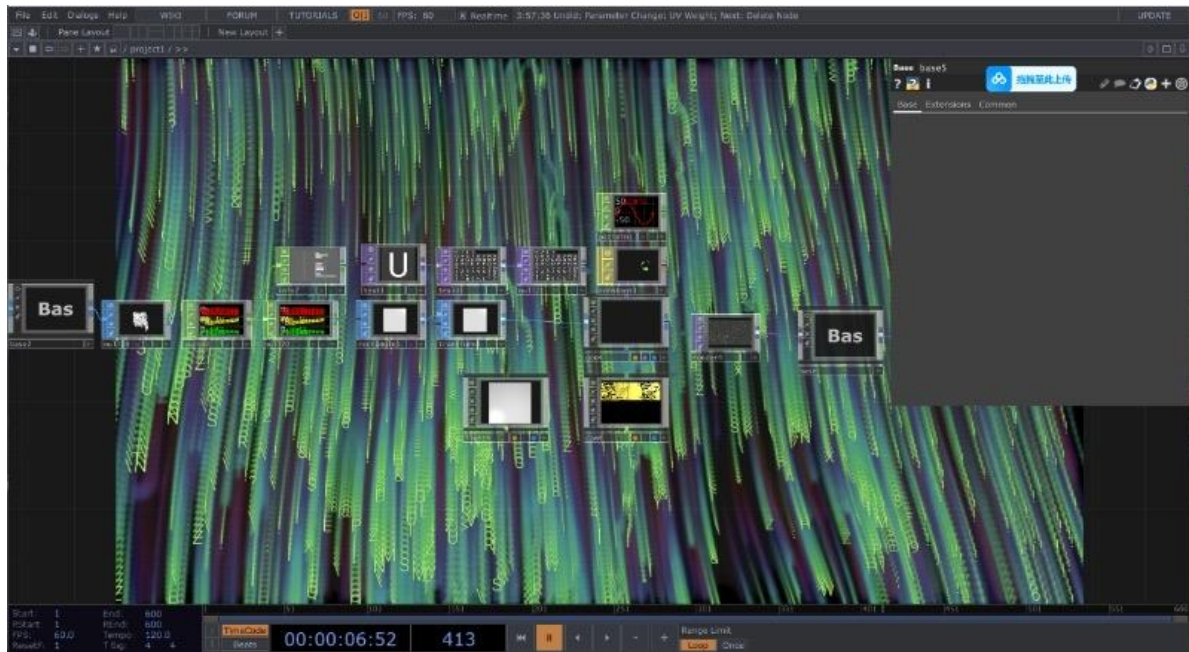


Figure 7: TouchDesigner scene network for the sin Envy

Emotional Logic

“Envy” is corrosive and isolating. Green, acid-like rain falls continuously through the space. A blurred figure stands alone in the darkness, slowly eroded by desire and silent observation.

System Behaviour

The Envy system operates through differential contrast and unstable deformation, transforming the depth silhouette into a volatile, fractured visual field. Kinect depth data is first segmented through a series of Threshold TOPs, isolating deformable regions within the figure. These segmented layers are routed into a composite chain built from Displace, Difference, and Noise TOPs, creating a continuously shifting surface that resists visual stability.

Two external drivers shape the behaviour of the scene: animated noise textures (generated by Noise and Turbulent Noise TOPs) and a flickering Ramp that simulates intrusive external illumination. Through a Difference TOP, the body is repeatedly compared against these variable inputs, producing tearing, flicker, and rapid tonal shifts characteristic of emotional disturbance. The system’s instability is not random—noise frequencies are tuned to create crack-like fracture patterns across the silhouette, rendered in high-saturation acid green through Lookup

and HSV adjustments.

A GLSL lookup shader adds jittering highlights that respond to depth peaks, making bright spikes travel across the body whenever the performer moves. This creates the sense that the figure is being pierced or irritated by external light sources—an embodiment of envy’s intrusive emotional dynamics. Because deformation updates at full frame rate, the slightest motion generates disproportionate visual agitation. The system therefore functions as a real-time instability engine: constantly comparing, constantly reacting, and never settling into equilibrium.

Motion Mapping

Envy is tuned to micro-movements. Minor shifts in posture trigger rapid visual trembling, making the body appear hyper-sensitive—always reacting, never at ease.

6.3 Wrath

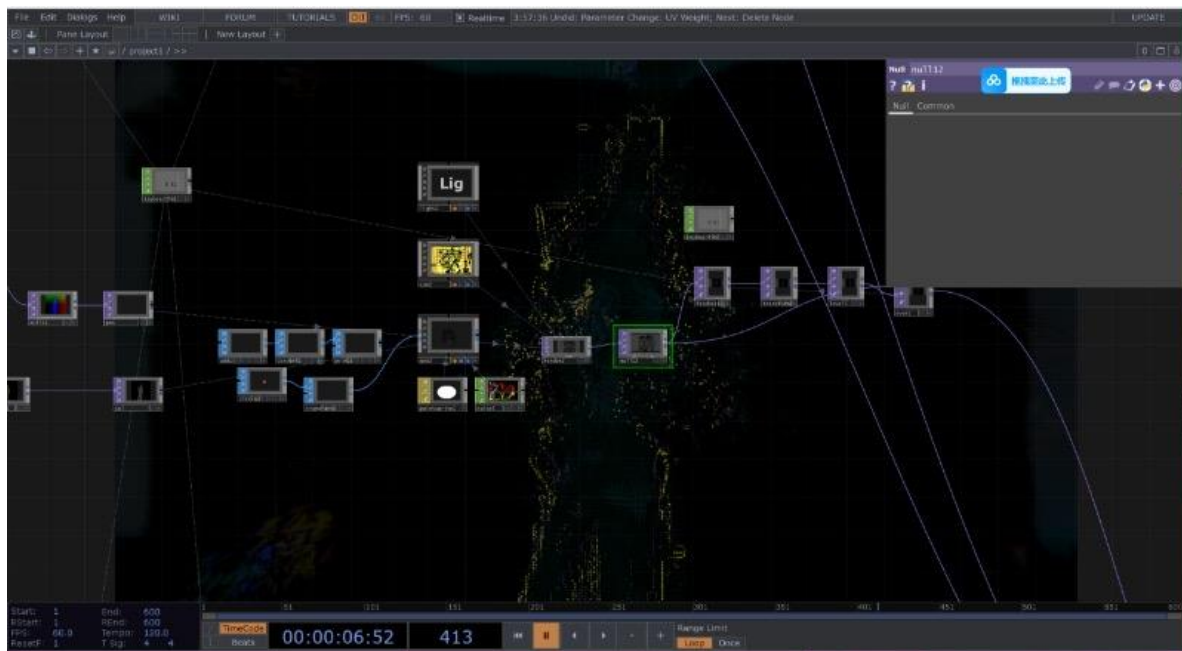


Figure 8: *TouchDesigner scene network for the sin Wrath*

Emotional Logic

“Wrath” embodies rupture, aggression, and explosive emotional release. Its behaviour is

intentionally unstable and volatile.

System Behaviour

Wrath is constructed around a velocity-driven distortion pipeline. Kinect skeleton tracking provides joint velocity values, which are processed using Math CHOP operations before being routed into Displace and colour-mapping networks within TouchDesigner. This ensures that physical acceleration directly modulates visual intensity.

The depth silhouette is isolated using a Threshold TOP (luminance threshold ≈ 0.097), establishing a clean base layer for aggressive processing. This silhouette then passes through an Edge Detect, Lookup, and Ramp chain to generate fractured red – green gradients that echo the high emotional temperature of anger. The Lookup TOP applies abrupt colour transitions that shift with movement intensity, giving the body a volatile chromatic signature.

A Feedback TOP accumulates frame data, turning rapid gestures into weapon-like afterimages that trail behind the figure. This feedback layer is intentionally unstable: without stabilisation filters, it responds immediately to motion, amplifying the sense of uncontrolled release. Displace TOPs apply jagged deformation proportional to the input velocity, causing the silhouette to rupture when the performer's motion spikes.

Gamma boosts and Level adjustments enhance edge contrast, making the body appear electrically overcharged. A GLSL filter further destabilises the output by modulating fragment brightness according to movement-derived parameters. Together, these processes create a system where technical instability becomes a material for representing emotional volatility: Wrath behaves like a controlled explosion, constantly on the edge of collapse.

Motion Mapping

The scene is highly velocity-sensitive. Quick movements cause distortion to spike instantly, transforming kinetic energy into visual violence.

6.4 Sloth – Technical Analysis

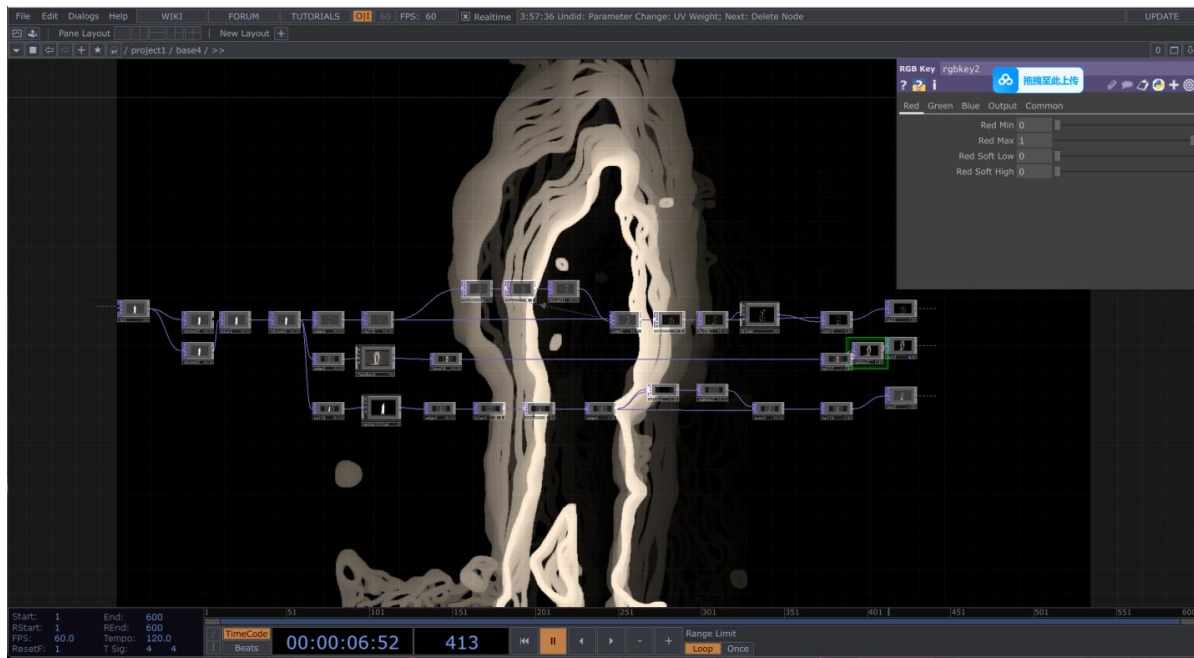


Figure 9: *TouchDesigner scene network for the sin Sloth*

Emotional Logic

“Sloth” is defined by inertia and delayed intention. The scene aims for heaviness and gradual dissolution.

System Behaviour

The Sloth system prioritises slow-reacting deformation built through an elongated image-processing chain. The Kinect silhouette is first softened through Blur and Erode TOPs, removing most fine detail. This blurred base image then travels through a horizontal sequence of Displace, Blur, and Noise operators that collectively introduce latency into the visual response.

Low-frequency Perlin noise fields modulate displacement, causing the body to drift and sag rather than shift sharply. These noise fields are tuned to produce gradual spreading rather than sudden peaks, reinforcing the impression of a body dissolving over time. Repeated blurring and displacement widen the silhouette, giving it a melted, softened appearance.

Unlike Wrath or Greed, Sloth deliberately avoids feedback-based accumulation, opting instead

for diffused delays. The long operator chain itself becomes a temporal filter: movement enters, but its influence is stretched, elongated, and dissipated before reaching the output. This structural delay is essential to the system's emotional tone; the visuals seem to lag behind the performer, evoking a sense of sluggishness and diminishing intention.

Motion Mapping

Sloth reacts best to very slow movement. Inputs accumulate with delay, making gestures feel weighted and sluggish, reinforcing a sense of minimal effort.

6.5 Greed – Technical Analysis



Figure 10: TouchDesigner scene network for the sin Greed

Emotional Logic

“Greed” explores accumulation and expansion, turning the body into a gravitational centre for digital “wealth”.

System Behaviour

Greed uses a tile-based displacement engine to visualise compulsive accumulation. Kinect depth

data is extracted and refined through Threshold and Level TOPs, forming a silhouette suitable for driving a field of tile units. These tiles—rendered as gold-toned geometric blocks—are generated via Tile and Tile Displace TOPs, with Noise modulation applied to create uneven, organic growth.

As the performer moves, the tile field rises, shifts, and stacks toward the silhouette, as though drawn by magnetic force. Noise fields of varying frequencies modulate the rate of expansion, producing areas of dense accumulation interspersed with thinner regions. Colour is handled through a Lookup + Ramp mapping that maintains a consistent gold palette, reinforcing the metaphor of wealth.

A Feedback TOP is introduced to generate persistent residue. When the performer moves, blocks appear to “stick,” forming layers that gradually thicken around the body. Transform TOPs shift these residues in subtle ways, making the accumulation appear active and spreading. The system is calibrated to ensure continuous growth: even moderate motion increases density, turning Greed into a self-multiplying visual engine.

Motion Mapping

Like Wrath, Greed is velocity-responsive, but instead of distortion, speed triggers expansion. Fast gestures produce dense layers of growth, mimicking compulsive hoarding.

6.6 Gluttony – Technical Analysis

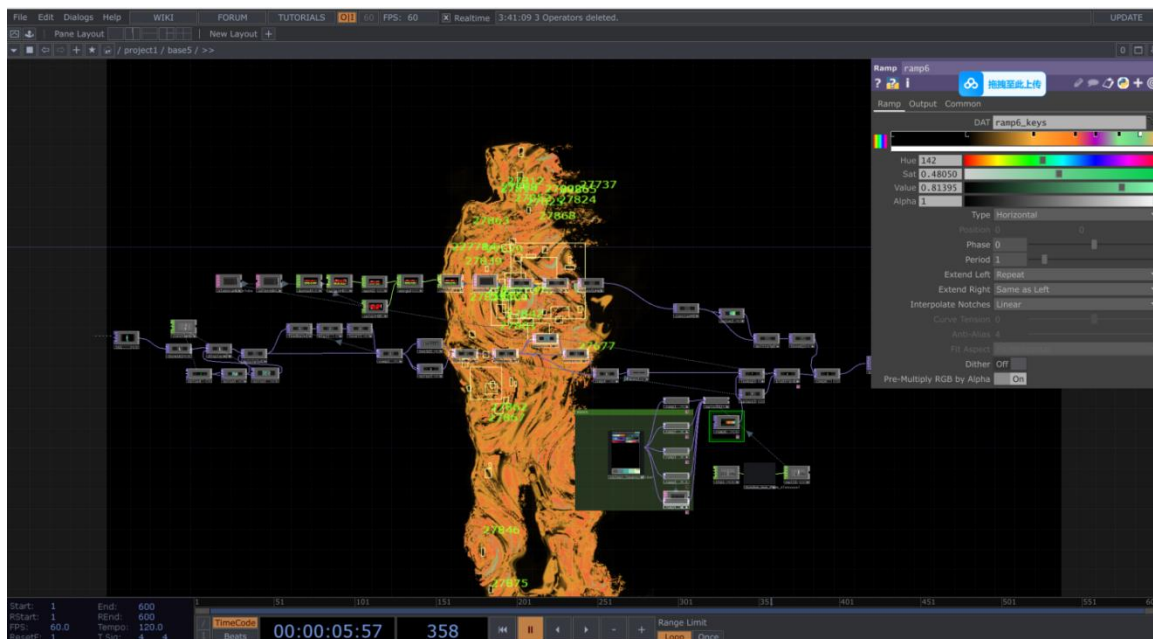


Figure 11: *TouchDesigner scene network for the sin Gluttony*

Emotional Logic

“Gluttony” revolves around excess, overflow, and swelling. The system depicts a body overwhelmed by its own expansion.

System Behaviour

The Gluttony system operates through noise-driven swelling and feedback-based accumulation. Kinect depth data is isolated via Threshold and Edge TOPs to produce a high-contrast contour suitable for deformation. This contour is then fed into Noise, Displace, and Blur TOPs, which generate fluid, bulbous distortions.

Animated Perlin noise drives both horizontal and vertical displacement, giving the body an unstable, inflated surface. The deformation is intentionally exaggerated, ensuring that large gestures produce visible expansion of the contour.

A Feedback TOP is crucial to Gluttony’s behaviour. By layering previous frames, the system produces thick, oily residues that cling to the silhouette, accumulating mass over time. Highlights are enhanced via luminance manipulation, making the inflated forms appear glossy and over-saturated. GLSL processing ensures the deformation updates smoothly in real-time, maintaining responsiveness while keeping the visual texture heavy and excessive.

Motion Mapping

The scene responds most strongly to large gestures, which cause rapid swelling and exaggerated expansion.

6.7 Lust – Technical Analysis



Figure 12: *TouchDesigner scene network for the sin Lust*

Emotional Logic

“Lust” evokes softness, ambiguity, and sensory attraction. The scene prioritises delicacy and fluidity rather than intensity.

System Behaviour

Lust is built around a softened image-processing pipeline that blends the silhouette into a pink – purple gradient field. After initial extraction, the depth silhouette undergoes blur and erosion operations to remove harsh edges. This softened form is then mapped through Noise, Ramp, and Lookup TOPs to create a gently shifting chromatic ambience.

Tile Displace TOPs driven by low-frequency Perlin noise introduce slow, wave-like undulations across the body. These oscillations are synchronised with a Timer CHOP, producing gradual pulses that reinforce the sensual, fluid tone of the scene. Layered alpha blending merges the figure with its surroundings, creating translucent overlaps that evoke atmospheric intimacy rather than solid structure.

A GLSL lookup shader adds faint proximity-based highlights, making the system respond to micro-movements with small glimmers that appear and fade smoothly. Processing latency is

slightly increased to avoid abrupt shifts, ensuring that all deformation flows with a slow, continuous quality.

Motion Mapping

Lust responds to micro-motion, translating subtle bodily shifts into faint highlights and soft pulsations.

6.8 Cross-Scene Interaction Logic

Although each sin has unique visual logic, all scenes derive from the same motion data. The system differentiates emotional behaviour through four primary mappings:

- Velocity → Wrath, Greed
- Micro-movement → Envy, Lust
- Stillness / centrality → Pride
- Slow motion → Sloth
- Large gestures → Gluttony

This mapping framework ensures that each emotional state expresses a distinct behavioural identity while remaining grounded in the same sensing pipeline.

7 Evaluation

7.1 User Testing & Evaluation

To assess the system’s usability, clarity, and expressive responsiveness, a user study was conducted with six participants from diverse backgrounds, including design students, programmers, and non-technical users. Each participant interacted with the installation for 5–10 minutes, during which their behaviour, interpretive responses, and technical difficulties were observed. Semi-structured interviews were then used to gather qualitative feedback. Across participants, the system was consistently described as visually distinctive and emotionally legible, but several issues emerged regarding sensitivity and clarity of interaction. Wrath and Greed were frequently reported as “too sensitive,” reacting strongly even to unintended or small gestures, indicating that their velocity thresholds were initially calibrated too low. Conversely, Sloth was often interpreted as unresponsive until participants discovered the intentional delay built into its design, suggesting that onboarding instructions must better communicate the logic of slow-response scenes.

Visual interpretation also varied across scenes. Users generally understood Gluttony’s swelling metaphor, though many suggested stronger contrast to make the visual accumulation more readable. Envy’s flickering textures were perceived as emotionally fitting but at times visually confusing, prompting refinements to noise amplitude and Difference-based tearing effects. An additional issue emerged during scene switching: without automatic feedback clearing, residual traces from Gluttony occasionally contaminated Pride, revealing the need for a system-level visual reset. Early prototypes also attempted to assign different gestures to each sin, but participants frequently triggered unintended switches during expressive movement, leading to the decision to adopt a single, deliberate fist gesture as the universal switching control. These findings informed targeted improvements, including revised velocity thresholds, clearer introductory instructions, enhanced lag cues for Sloth, and optimized feedback-reset mechanisms, collectively strengthening the clarity and reliability of the interaction experience.



Figure 13: *Photograph of Seven Sins installation and hardware setup*

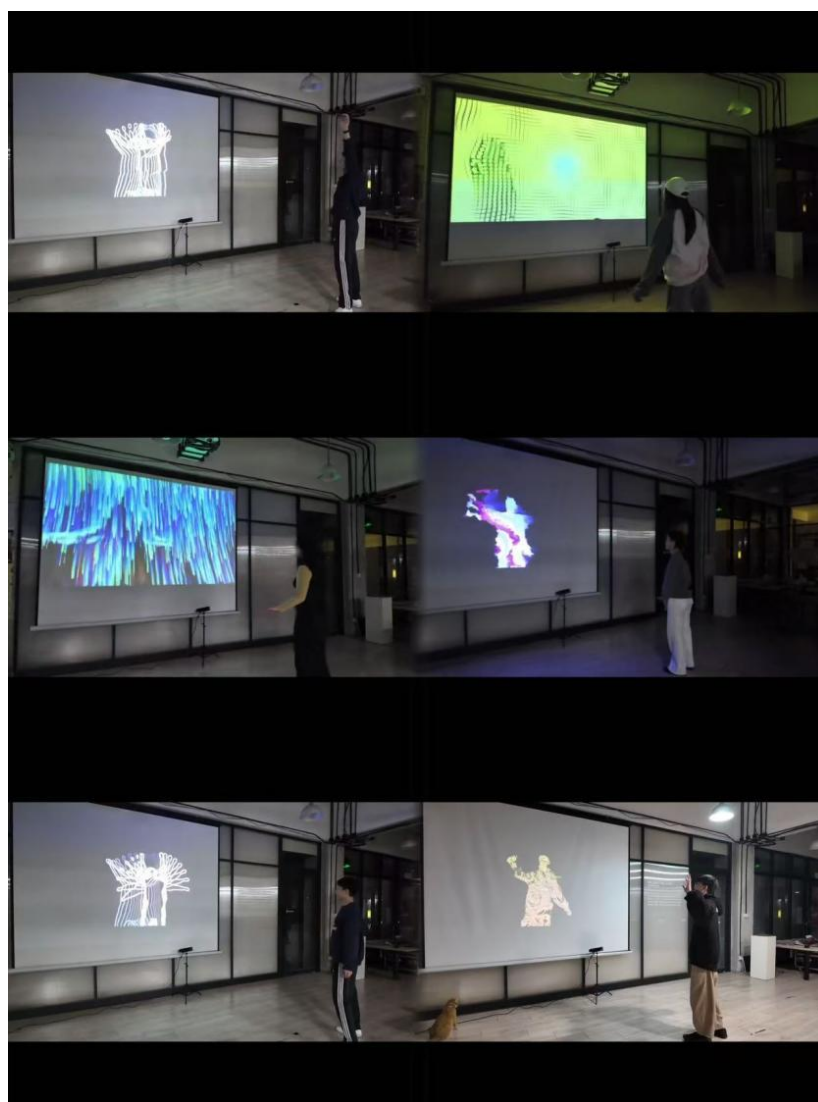


Figure 14: *Photograph of participants interacting with the Seven Sins installation*

Video: <https://youtu.be/KYAyKzq-GE8?si=Feh0Q-7l2aiWqxXm>

7.2 Limitations

Although *Seven Sins* demonstrates the potential of real-time generative visuals to construct emotional meaning from bodily motion, a number of limitations remain at both technical and conceptual levels.

First, the system operates almost entirely within the visual domain. While this decision allowed the project to focus specifically on image-based emotional construction, it also limits the overall sensory depth of the experience. In embodied perception, emotion is rarely produced through vision alone, but rather emerges through the interaction of multiple sensory channels, including sound, touch, and spatial atmosphere. By remaining primarily visual, the current system frames emotion largely as something to be seen, rather than something fully felt through the body.

Second, the emotional logics assigned to each of the seven sins are manually designed rather than learned from data. This approach aligns with the project's critical position against recognition-based emotion models, yet it inevitably embeds the author's subjective interpretation within the system. The behaviours of each sin are not discovered through the participants' movements, but are instead shaped through pre-authored aesthetic and computational decisions. As a result, the system does not reveal an "objective" emotional structure, but rather a constructed one, shaped by cultural references, artistic intention, and algorithmic design. This raises broader questions concerning authorship, cultural bias, and the politics of emotional coding within generative systems.

Third, the scope of the user testing was limited in both scale and demographic diversity. While informal testing played an important role in refining gesture stability, sensitivity thresholds, and scene transitions, these observations remain primarily qualitative. Given that the interpretation of emotion is strongly influenced by personal experience, cultural background, and media literacy, the small sample size restricts the extent to which the findings can be generalised.

Finally, the system is constrained by the technical limitations of the Kinect v2 sensor. While the device performs reliably in capturing full-body movement and large-scale gestures, it lacks sufficient precision for subtle finger articulation and fine-grained micro-movements. This limitation shaped the interaction design toward relatively coarse bodily actions, and restricted the system's capacity to respond to more delicate emotional nuances.

7.3 Future Work

Several directions for future development emerge from both the technical outcomes and the conceptual framework of this project.

From a technical perspective, future iterations could explore the integration of machine learning methods for motion clustering or affective pattern formation. Rather than replacing the current rule-based system with recognition models, such methods could be used to introduce adaptive or evolving emotional behaviours that shift over time in response to accumulated movement data. This would allow the emotional system to remain generative while gaining an additional temporal and developmental dimension.

At the sensory level, the project could be expanded into a more fully multimodal installation. The addition of generative sound, spatial audio, haptic feedback, or programmable lighting could transform the emotional experience from a primarily visual encounter into a deeply embodied and immersive environment. Such extensions would allow emotion to be felt across the entire perceptual field, rather than being confined to visual deformation alone.

On a theoretical and methodological level, future research could involve cross-cultural audience studies to examine how different cultural contexts interpret the emotional behaviours of the seven sins. This would provide valuable insight into the cultural specificity of computational emotion systems and contribute to wider debates on digital affect, embodiment, and generative aesthetics.

Taken together, these directions suggest that Seven Sins functions not only as an interactive artwork, but also as an experimental research platform for investigating how computation participates in the construction of emotion, how the body is reconfigured through algorithmic systems, and how human self-perception shifts through sustained interaction with generative environments.

8 Conclusion

The results of this project demonstrate that generative visual systems can construct emotional meaning not through affect classification but through the transformation of bodily motion into algorithmic behaviour. The Seven Sins system shows that the emotional identity of an interactive environment can emerge from the relationships between depth signals, motion patterns, and real-time visual processes. Rather than detecting whether a participant is angry, calm, or excited, the system defines emotion as an aesthetic and behavioural field shaped by how the body moves. This approach reframes emotional computing away from recognition and toward expressive construction.

From the perspective of interaction design, the project reveals that emotional clarity depends not only on visual aesthetics but also on the legibility of motion – response mappings. User testing confirmed that each sin’s visual logic—velocity-driven explosions, micro-movement flicker, delayed deformation, or center-pose amplification—was generally interpretable as an emotional behaviour, but sensitivity calibration played a crucial role. The transition from multiple gestures to a single deliberate fist gesture for scene switching further highlighted the technical and experiential importance of minimizing accidental triggers in embodied interaction.

Conceptually, the project contributes to broader discussions on digital embodiment by showing how the body becomes both source and material for emotional generation. The digital figure is not a passive reflection of the participant but an altered, expanded, and affectively responsive double. This resonates with theories of the posthuman body, where identity is distributed across biological and computational processes. Through real-time deformation, feedback accumulation, and motion-dependent behaviour, the system articulates emotion as something co-created between human and algorithmic agency.

Technically, the project demonstrates the viability of a multi-scene, GPU-accelerated architecture in TouchDesigner, capable of sustaining simultaneous generative pipelines at interactive frame rates. The study also highlights the challenges of motion stability, parameter tuning, and visual residue management — factors that become central when emotion is constructed through dynamic image systems. These findings point toward future research directions in adaptive emotional systems, multimodal affect synthesis, and culturally grounded interpretation of generative emotions.

Overall, *Seven Sins* argues for a shift in how emotion is imagined in computational art: not as something to be detected, but as something to be produced, shaped, and negotiated through embodied interaction. The digital body becomes an experimental site where emotional behaviour can be recomposed and reinterpreted, opening new possibilities for affective computing, creative AI systems, and interactive digital art.

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